



Evaluating the Impact of Immersive Technologies on STEM Education in Sri Lanka's Secondary and Tertiary Curriculum

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Feasibility Study

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1. Executive Summary

This feasibility study evaluates the development and implementation of a specialized Augmented Reality (AR) infrastructure designed to modernize Information Technology (IT) education within the Sri Lankan secondary school system. While the national curriculum is well-structured, IT instruction currently relies on static illustrations and theoretical explanations to describe complex, "invisible" digital processes, which often leads to rote memorization rather than practical understanding. The primary objective of this component is to develop a robust mobile AR application that allows students to visualize abstract IT concepts such as data flow, network topologies, and computing logic by overlaying interactive 3D simulations directly onto their standard government-issued textbooks.

The proposed technical solution utilizes a smartphone camera to scan existing 2D textbook diagrams, triggering dynamic 3D models and animations through the integration of Unity and Vuforia's image-target recognition. By transforming flat diagrams of network structures or hardware components into rotatable, interactive simulations, the platform shifts the learning process from passive observation to active discovery. This approach is specifically designed for high accessibility in the local context, leveraging the mobile devices students already own and integrating seamlessly with physical textbooks without requiring expensive laboratory infrastructure or specialized equipment.

A detailed feasibility assessment confirms the technical viability and operational suitability of the system. Technical analysis proves that by using real-time 3D rendering optimization, high-quality educational content can be delivered effectively on mid-range Android devices, which is critical for widespread adoption in Sri Lankan classrooms and for functioning offline. Market evaluation highlights a significant research gap in localized AR tools for Grade 10 students, indicating a strong demand for innovative digital pedagogy that supports conceptual mastery over surface-level learning.

The positive findings of this study support the advancement of the IT component as a cornerstone of the broader STEM learning module. By making fundamental computing principles tangible and accessible, the application has the potential to transform IT education from a theoretical subject into an experiential one. This project addresses critical local challenges in resource-constrained environments while aligning with global trends in interactive media, positioning it as a scalable and impactful tool for modernizing digital literacy in Sri Lanka.

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2. Introduction

2.1 Background

Information Technology (IT) education is vital for preparing students for a digital future; however, in Sri Lanka, the subject is often taught using old fashioned methods that rely on static 2D images in textbooks. While these methods cover the syllabus, they are not enough to help students understand dynamic or "invisible" technical processes. This challenge is very clear in the Grade 10 IT curriculum, where students must learn about data flow, network topologies, and computing logic. These concepts involve movement and interaction that cannot be easily shown through text or flat drawings, often leading students to memorize facts for exams instead of gaining a real, practical understanding.

In the IT component specifically, students struggle to visualize how digital systems work, such as how data packets move through a network or how software logic flows. Without interactive tools, these topics remain abstract and difficult to grasp. In many Sri Lankan schools, the lack of modern computer labs and digital equipment further limits students' chances to see these processes in action. Teachers also find it difficult to explain complex technical structures using only a blackboard or limited resources. Currently, no major research has focused on building an Augmented Reality (AR) tool specifically designed to work with the existing Sri Lanka's government issued IT textbooks for these grade levels.

Augmented Reality (AR) offers a powerful way to solve these problems by allowing students to see and interact with 3D simulations directly on their textbook pages. Using a common smartphone or tablet, a student can scan a diagram such as a network structure and see it "come to life" as a 3D model that can be rotated and explored. This turns passive reading into an active process of discovery. AR makes abstract IT concepts tangible while there remaining a low cost and scalable solution that does not require schools to buy expensive new hardware. By combining AR technology with the national IT curriculum, we can modernize how technology is taught, helping students build deep conceptual mastery and practical skills without needing high-end laboratory infrastructure.

2.2 Objective

- **Technical tools** -Check the hardware and software needed, like Unity and Vuforia, to make sure the AR app works well on mobile phones
- **Check image scanning** - Make sure the app can easily recognize and scan 2D diagrams from the Grade 10 IT textbooks.
- **Select IT topics** -Pick difficult IT concepts, like how data moves or how networks are built, that are easier to learn using 3D models.
- **Optimize for local phones** - Ensure the app runs smoothly and stays fast on the mid-range Android phones that most Sri Lankan students use.
- **Build accurate models** -Create correct 3D models of computer parts and digital processes so the lessons are technically right.
- **Design for easy use** -Create a simple app design (UI/UX) so that both students and teachers can use it in class without trouble.

2.3 Scope

- **Technical Feasibility** - Testing the right phones and software (like Unity and Vuforia) to make sure the app scans pictures and shows 3D models smoothly.
- **Market Feasibility** - Looking at how Grade 10 students and teachers can use AR to understand "hidden" IT topics like data flow and network structures.
- **Financial Feasibility** - Planning the budget and finding cheap ways to keep the app running in local schools.
- **Operational Feasibility** - Making sure the app is easy to use with regular school textbooks and ensuring it works without the internet.
- **Environmental and Ethical Feasibility** - Using digital models instead of physical equipment to save resources and making sure all IT lessons are accurate and safe for students.

3. Problem Statement

STEM education in Sri Lanka, particularly in the field of Information Technology, faces major challenges in teaching abstract and dynamic processes through traditional textbooks. Current IT lessons for Grade 10 students rely heavily on flat diagrams and text to explain complex digital systems. These conventional methods often fail to show how "invisible" processes work, such as the flow of data packets in a network or the internal logic of a computer system. Because these concepts involve movement and three-dimensional relationships, static images cannot accurately demonstrate how they function in the real world.

Current teaching methods lack interactive visualization, which limits a student's ability to connect theory with practical IT skills. As a result, students often resort to rote memorization to pass exams rather than gaining a deep understanding of how technology works. Misconceptions are common because students cannot observe or manipulate the abstract structures they are reading about. There is a clear and urgent need for a learning system that provides real-time, 3D visualization to make these "invisible" IT concepts tangible and easier to explore.

Augmented Reality (AR) technology can bridge this gap by overlaying interactive 3D simulations directly onto the diagrams in government-issued IT textbooks. By using a smartphone to scan a page, students can observe and interact with models of network topologies or computing logic in a safe and low-cost environment. The proposed AR platform offers an accessible way to modernize IT education, helping students in Sri Lankan classrooms move from passive learning to active discovery and better conceptual understanding.

4. Project Description

The proposed AR-based educational system uses real-time 3D simulations to create an interactive learning experience specifically for Information Technology students. As the Technology Lead, this component focuses on using detailed 3D models that allow students to see and control abstract digital processes, such as data flow, network structures, and computing logic. These real-time interactions let students explore how data moves through a system over time, helping them visualize complex relationships and understand the cause-and-effect dynamics of digital communication.

A key feature of the system is its ability to place interactive 3D IT models directly on top of 2D textbook diagrams using mobile devices and Vuforia image-target recognition. Students can rotate the models, change parameters like adding or removing a node in a network and observe the simulation from different angles to make the concept clearer. The system includes guided modules that follow the national IT curriculum for Grades 10, offering structured lessons and interactive feedback that allow students to practice and explore at their own pace.

By combining 3D visualization, interactivity, and direct alignment with school textbooks, this AR platform provides a safe, low-cost, and exciting way to learn IT. This system has the power to change how IT is taught in Sri Lankan classrooms by making "invisible" technical concepts easy to see, increasing student interest, and helping teachers deliver high-quality technical lessons even in schools with limited computer lab facilities.

5. Feasibility Analysis

5.1 Technical Feasibility

The technical check shows that this project can be done using standard tools and mid-range mobile phones.

- **Software Tools** - We will use **Unity** for building the app and **Vuforia** for scanning textbook pictures.
- **3D Design** - We will use **Blender** to make 3D models of computer parts.
- **Hardware** - The app is designed to work on regular **Android phones** with cameras, so students can use it easily.
- **Main Stages** - The work includes making the 3D models, connecting them to the AR app, and testing them in real classrooms.
- **Technical Goals** - The main goal is to make sure the 3D animations run smoothly on cheaper phones and work **offline** without the internet.
- **Testing** - We will build a prototype to test how well the app recognizes pictures and get feedback from teachers and students.

5.2 Market Feasibility

The market study shows a strong need for new way to teach Information Technology in Sri Lankan schools, specifically focusing on Grade 6–9 students.

- **Target Audience** - The primary users are secondary school students and IT teachers in Sri Lanka who want better ways to learn STEM subjects.
- **Market Needs & Trends** - There is a growing demand for digital tools that follow the local school syllabus and reduce the need for memorizing facts.
- **Competitive Analysis** - Most current educational apps use only flat 2D pictures and have very little interaction. Our AR platform is unique because it offers hands-on 3D learning.
- **Specific Gap** - Currently, there is no major AR research or app that focuses on government-issued textbooks for IT.
- **Evidence of Demand** - Early trials and surveys show that both students and teachers are very interested in using visual and interactive tools for learning.

5.3 Financial Feasibility

- **Cost Estimates** - Estimated development cost is Rs. 20,000–30,000, covering AR app development, 3D content creation, and necessary hardware (Android devices, software licenses).

5.4 Operational Feasibility

- **Team Requirements** - We need a small team including AR developers to build the app, UI/UX designers to make it easy to use, and 3D animators to create the IT models.
- **Easy to Use** - The app is designed to work with existing school textbooks, so teachers do not need to change how they teach.
- **Works Anywhere** - The app can be used in both city and village schools because it works on regular mobile phones and does not need the internet to function.
- **Simple Training** - We will provide basic training for teachers and students, so they will feel comfortable using the AR tools during lessons.
- **Regular Updates** - We will use feedback from students to keep the app updated and make sure it stays helpful for a long time.

5.5 Environmental or Ethical Feasibility

- **Environmental Impact** - Because the AR app is fully digital, it has almost no negative impact on the environment. It allows students to explore IT concepts without needing physical lab setups or electronic waste from old hardware.
- **Sustainability** - The app replaces resource-heavy experiments with digital simulations, promoting a low-cost and eco-friendly way to learn about technology.
- **Ethical Standards** - All IT content is checked by educators to make sure it is accurate and follows the national curriculum.
- **Inclusivity and Privacy** - The system is designed to be inclusive for all students and strictly protects student data privacy to ensure a safe digital learning environment.

6. Risk Analysis

- **Device Compatibility** -The Augmented Reality (AR) application must be optimized to run smoothly on a wide variety of Android devices, ranging from older models to modern smartphones.
- **Software Stability and Performance** - It is critical to ensure the app is free of bugs and can maintain high-quality, real-time 3D rendering of complex IT processes without lagging.
- **Image Recognition Reliability** - Using Vuforia, the app must consistently recognize 2D textbook diagrams under various lighting conditions to trigger the 3D models.
- **Technical Accuracy** - IT simulations such as data flow and network structures must be technically accurate to ensure they correctly represent computing principles.

2. Financial Risks

- **Initial Development Costs** -Funding is required for specialized AR app development, high-quality 3D modeling, and software licensing fees.
- **Adoption Constraints** - Some schools may be slow to adopt the technology due to limited budgets or a lack of familiarity with digital learning tools.

3. Operational Risks

- **User Training and Support** - Teachers and students may need initial guidance and training to feel confident using the AR tools during lessons.
- **Maintenance and Updates** - The application will require regular technical support, troubleshooting, and updates to stay compatible with new mobile operating systems.
- **Scalability Challenges** - Expanding the IT modules to include more advanced topics or additional grade levels will require more technical resources and expertise.

7. Recommendations

- **Work with Experts** - Collaborate with IT teachers and curriculum experts early on to make sure the 3D models and IT lessons are correct.
- **Build a Prototype** - Create a test version of the app to check how well it scans textbook pictures and to fix any technical bugs.
- **Get Student Feedback** - Test the app with students to see if it is easy to use and if it helps them understand IT better.
- **Focus on Speed** - Make sure the 3D animations stay smooth and fast, even on older or cheaper mobile phones.

8. Conclusion

The proposed Augmented Reality (AR) system for Information Technology (IT) education is a practical and innovative solution for secondary schools in Sri Lanka. By using AR, the platform allows students to finally "see" and interact with difficult, abstract IT concepts such as data flow, network structures, and computing logic that were previously limited to flat pictures in a textbook. This transition from passive reading to active 3D exploration helps improve student understanding and keeps them more engaged in the learning process.

The feasibility study confirms that this IT component is technically achievable using standard mobile devices and works effectively in local classroom environments without needing the internet. It offers a low-cost way to modernize STEM education, making high-quality technical learning accessible to all students regardless of their school's physical lab facilities. Ultimately, this AR platform has the potential to transform traditional textbook learning into an immersive experience, helping students build the real-world digital skills they need for the future.

9. Reference

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